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Electron diffraction with bound electrons: The structure sensitivity of Rydberg Fingerprint Spectroscopy

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ABSTRACT

Rydberg Fingerprint Spectroscopy (RFS) has proven to be a powerful method to probe molecular structures, especially transient structures with large amounts of internal energy. The technique is related to electron diffraction in that the molecular structure sensitivity originates from the wavefunction phase shift of the probe electron induced by the charge distributions in the molecule. Exploiting the close relationship between the two techniques, we investigate the origin of the molecular structure sensitivity of RFS by introducing a molecular ion core model that is solved analytically using perturbation theory as well as numerically using a numerical-grid method. The dependence of the Rydberg electron energy on some molecular parameters is investigated and the effectiveness of the numerical-grid method in solving our one-electron Schrödinger equation is validated. A reasonable consistency between experimental and computed quantum defect values is shown for specific molecular ion core model parameters.

1. Introduction

The identification of transient molecular structures, i.e. structures that exist for brief periods of time during chemical reactions, is an important goal in our quest for deeper insights into chemistry [1,2]. While inherently static structure determination methods have been adapted to fast time scales [3–6], often there are severe limitations. Time-resolved electron diffraction of gas phase molecules in particular suffers from the conflicting needs for high signal to noise ratios and the necessity to keep the number of electrons in a pulse low to avoid space charge interactions [7–9]. While successes have been achieved with such methods, and while it is possible to reduce the space charge limitations by using highly relativistic electrons [10], the development of new structural methods that are inherently better suited for ultrafast time resolution remains an important goal.

We have recently introduced Rydberg Fingerprint Spectroscopy (RFS) [11–13], a structure-sensitive spectroscopic technique that has many unique advantages. The method can be applied equally to all large and small molecules as long as they have a positive ion core [14]. Implemented with ultrafast pulsed lasers, the method is inherently able to time-resolve ultrafast structural processes [15]. Further, and unique among the structural techniques, RFS senses the global molecular structure, including isomeric and conformeric forms [16,17]. Because Rydberg Fingerprint spectra are insensitive toward vibrational excitation, they are not affected by high sample temperature [18]. Finally, the complexity of the spectra does not scale with the size of the molecule, so that spectra of even large molecules are easy to assign. However, to date the spectra cannot be inverted to deduce a molecular structure, restricting the technique to a fingerprint analysis rather than a structure determination method.

RFS measures the energy of electrons in molecular Rydberg states. Experimentally, this can be achieved by exciting an electron from the ground state to a Rydberg excited state, and then measuring the photoionization photoelectron spectrum. For principal quantum numbers $n \geqslant 3$ the Rydberg orbitals are similar to those of the hydrogen atom [19]. But in molecules, the Rydberg electron occasionally penetrates and scatters off from the molecular ion core, where it experiences a more negative potential than in the hydrogen atom, leading to an increase in its kinetic energy and hence the radial oscillation frequency. As a result, the nodes of the radial wavefunctions are drawn nearer to the molecular center than in the hydrogen atom [20]. The phase shift is mirrored by a shift in the Rydberg electron energy as is illustrated in Fig. 1.

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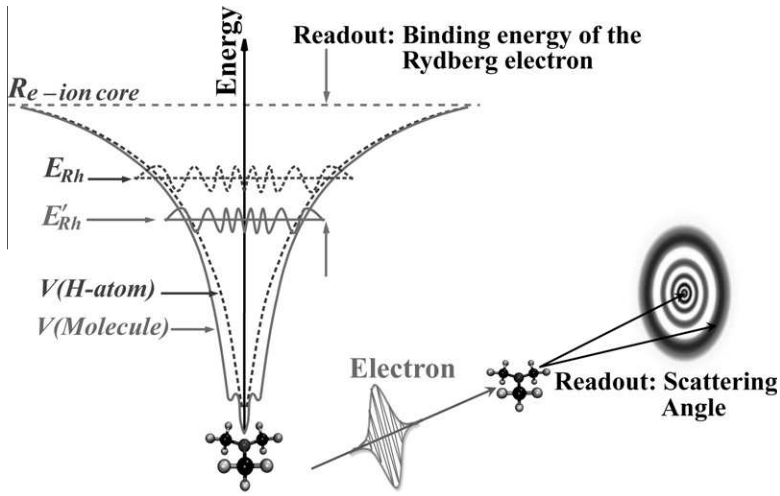

Fig. 1. Relationship between Rydberg Fingerprint Spectroscopy and electron diffraction. An illustration of a Rydberg state in the trimethylamine (TMA) molecule. The Rydberg electron penetrates the ion core, where the potential is lower than that of a hydrogen atom. The wavelength of the radial wavefunction inside the ion core is decreased, introducing a phase shift. The structure sensitivity of electron diffraction arises from the phase shift introduced on a traveling wavefunction by the molecular charges.

To a good approximation, the energy of an electron in a Rydberg state, resembling that of the hydrogen atom, is given by

$$E = -\frac{R_h}{(n-\delta_l)^2} \tag{1.1}$$

where R_h is the Rydberg constant (13.606 eV) and δ_l is a constant called quantum defect that depends on the orbital angular momentum [21,22]. Since the Rydberg electron energy shift comes from the molecular potential downshift, it is sensitively dependent on the positions of atoms in the molecule, as well as the chemical bonds between them. An example of such structural sensitivity is shown in Fig. 1, in which the characteristic molecular potential of trimethylamine (TMA) molecule is portrayed schematically. Generally speaking, s orbitals ($l = 0$) have quantum defects close to 1, while p orbitals ($l = 1$) have δ_l values between 0.3 and 0.5 and d orbitals ($l = 2$) have values less than 0.1 [23].

From this description it follows that the structure sensitivity of RFS arises from the very same phase shifts that are responsible for the structure sensitivity of electron diffraction. In both cases, the wavefunction of the probe electron is perturbed in a systematic fashion by the charge distributions of the particles in the molecule. The difference is in the readout: in diffraction, the phase shifts are projected onto the detector and observed as an interference pattern, whereas in RFS the readout is through the Rydberg electron energy. In most cases, electron diffraction probes molecules in their ground state structures, while RFS probes the structure in a Rydberg state, which most often closely resembles that of the ion. It is this connection between RFS and electron diffraction that we wish to explore in further detail here.

The relevant theory to describe Rydberg states is the Quantum Defect Theory [24], which in modern numerical calculations is usually implemented as Multichannel Quantum Defect Theory (MQDT) [25]. These theories not only explain the dependence of the quantum defect on the molecular potentials as well as the connection between the Rydberg electron wavefunction phase shift and the quantum defect [26], but also open the door to the accurate *ab initio* calculation of the quantum defect in atoms [27–30] and simple molecules [21,31–36]. Nevertheless, the complexity of MQDT confines it to small molecular systems, so that the broad connections between Rydberg electron energies and molecular properties, such as size, shape and charge distribution, are not well understood, motivating the search for a complementary approach.

2. Rydberg Fingerprint Spectroscopy and electron diffraction

One of the goals of our present work is to expose the close relationship between RFS and electron diffraction. It is fortunate that the changes of the Rydberg electron energies with molecular shape are large. One may therefore be justified in treating the Rydberg electron as a single electron orbiting a molecular ion core in the potential created by the charge distribution of the electrons and nuclei of the ion core. The energy of the Rydberg electron is then given as

$$E = \langle \psi \mid \hat{T} + \hat{V} \mid \psi \rangle \tag{2.1}$$

where $\hat{T}$ and $\hat{V}$ are the kinetic and potential energy operators, and ψ is the total molecular wavefunction. Separating the latter into a product of the wavefunctions of the Rydberg electron and the remaining ion core, $\psi = \psi_{Rh}\psi_{M^+}$, one can write the energy of the Rydberg electron as

$$E = \langle \psi_{Rh} \mid \hat{T} + \langle \psi_{M^+} \mid \hat{V} \mid \psi_{M^+} \rangle \psi_{Rh} \rangle \tag{2.2}$$

The potential energy term results from the interaction of the Rydberg electron with all the charges of the molecular ion core, i.e. the nuclei with charge Z_α at distance $r_{n,\alpha}$ and the electrons i at distance $r_{e,i}$ (measured from the Rydberg electron).

$$\hat{V} = -\sum_\alpha \frac{1}{r_{n,\alpha}} \cdot Z_\alpha + \sum_i \frac{1}{r_{e,i}} \tag{2.3}$$

Turning now to electron diffraction, the elastic scattering of an electron from a molecule in state ψ_M is defined by the scattering amplitude [37,38]

$$g_{i,i} = \langle \psi_M \mid \hat{L} \mid \psi_M \rangle \tag{2.4}$$

with the electron scattering operator

$$\hat{L} = -\sum_\alpha Z_\alpha \cdot e^{i\vec{S}\cdot\vec{r}_{n,\alpha}} + \sum_i e^{i\vec{S}\cdot\vec{r}_{e,i}} \tag{2.5}$$

Framed in this way, the similarities and differences of electron diffraction and RFS become apparent (Fig. 1). The structure sensitivity for both techniques derives from the interaction of one electron, which is freely moving in diffraction but bound in RFS by all nuclear and electron charges of the molecule or ion core. The main difference, as we noted earlier, is in the readout of the interaction:

in electron diffraction one projects a diffraction pattern on a detector, giving rise to the customary $e^{i\vec{S}\cdot\vec{r}}$ terms in the scattering operator. In RFS, the readout via the electron energy gives rise to the Coulombic $1/r$ term.

This suggests that like in electron diffraction, it may be possible to use an independent-atom approach for the analysis of RFS. Such a treatment remains to be developed and validated. An alternative approach, however, is to assume that the exact molecular ion charge distribution defines a fixed potential in which the Rydberg electron moves. Such an approximation could be implemented by using standard *ab initio* methods to calculate the charge distribution in a molecule of interest. The wavefunction and energy of the Rydberg electron could then be calculated by solving its one-electron Schrödinger equation.

In the present work, our first goal is to illustrate the origins of the structure sensitivity using a molecular ion core model potential that can be treated analytically using perturbation theory. Secondly, we use a numerical partial differential equation solver to calculate the one-electron energies for the same model systems. The results of those calculations nicely generalize trends in the perturbation results, and show a reasonable consistency between experimental and computed quantum defect δ_l values for specific molecular ion core model parameters.

3. A spherical model

3.1. Model construction

To construct a reasonable model for RFS, we first notice that the size of a molecular ion core is much larger than the hydrogen atom core. We therefore take the geometry of the model as a sphere with a finite radius σ. Within this core, the molecular Rydberg electron is simultaneously repelled by the inner electrons and attracted to the positive nuclei. It is only while the Rydberg electron is far away from the molecular ion core that the potential energy it experiences equals that of an electron in the Coulomb potential of a single positive charge. Inspired by the treatment of the shielding effect in multi-electron atoms, we construct the molecular ion core model with an effective total nuclear charge of +1 derived from a positive charge of magnitude Z ($Z > 1$) placed at the center of the sphere and negative charges of magnitude $Z - 1$ uniformly distributed throughout the sphere (inset in Fig. 2). In the limit of $Z = 1$, this model becomes identical to the hydrogen atom. In the limit that $\sigma \to 0$, the full positive charge is uniformly distributed throughout the core.

3.2. Electric potential for the molecular ion core model

Regardless of the value of Z, the total integrated charge density is a constant

$$\int d\vec{r}\rho(\vec{r}) = 1 \tag{3.1}$$

where the vector $\vec{r}$ is the distance between the Rydberg electron and the center of the sphere. The resulting potential can always be written as

$$V(\vec{r}) = -\int d\vec{r}'\rho(\vec{r}')\frac{1}{|\vec{r}-\vec{r}'|} \tag{3.2}$$

In the uniform core limit, the molecular ion core potential is just [39]

$$V(\vec{r}) = \begin{cases} -\frac{1}{2\sigma}\left(3 - \frac{r^2}{\sigma^2}\right) & (r \leqslant \sigma) \\ -\frac{1}{r} & (r > \sigma) \end{cases} \tag{3.3}$$

indicating that the Rydberg electron experiences an electric potential similar to that of a single harmonic oscillator inside the molecular ion core.

For the more general case of our ion core model, the electric potential comes from two independent charge sources: the central positive charge and the uniformly distributed negative charge. The charge density distribution in the latter part gives

$$\int d\vec{r}\rho(\vec{r}) = 1 - Z \tag{3.4}$$

Therefore, the potential resulting from the latter source is just Eq. (3.3) multiplied by $(1 - Z)$. The central positive charge contributes a simple Coulomb potential $-(Z/r)$. Therefore, the total electric potential is just

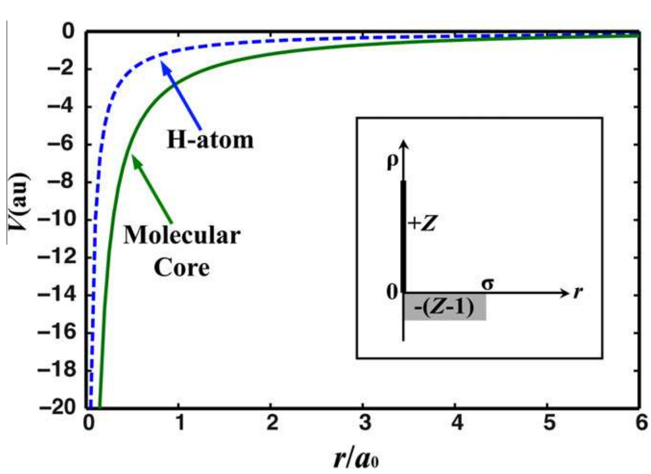

Fig. 2. Scheme of the molecular ion core model and its potential, in comparison to the hydrogenic Coulomb potential. The charge distribution in the spherical core region is shown in the inset. Positive charges Z, with $Z > 1$, are positioned at the core center while $Z - 1$ negative charges are uniformly distributed elsewhere in the core of radius σ. In the scheme $Z = 3$ and $\sigma = 10$.

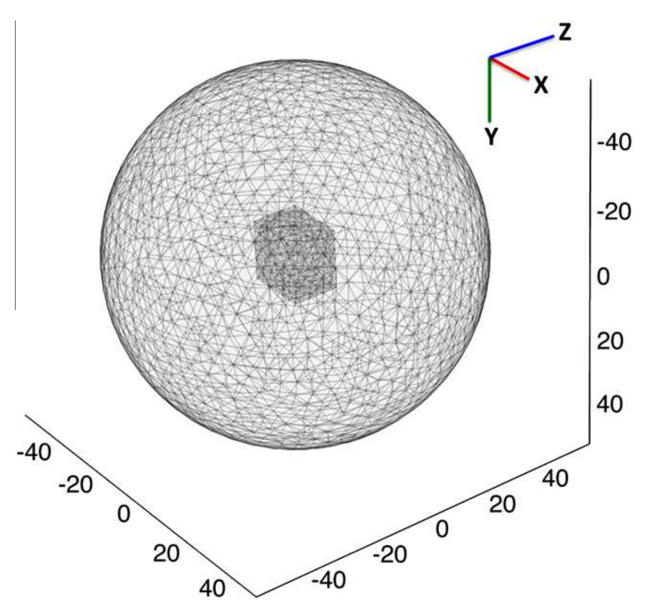

Fig. 3. The boundary geometry and meshed grids for solving the Schödinger equation in the numerical software package. The outer sphere has a radius of 50 Bohr (a_0) while the inner block, with a finer mesh, measures $17a_0{*}17a_0{*}13a_0$. The grids are meshed to "finer" level as predefined in the software.

$$V(\vec{r}) = \begin{cases} -\frac{Z}{r} + \frac{(Z-1)}{2\sigma}\left(3 - \frac{r^2}{\sigma^2}\right) & (r \leqslant \sigma) \\ -\frac{1}{r} & (r > \sigma) \end{cases} \tag{3.5}$$

As is shown in Fig. 2, inside the core the model potential is more negative than that of the hydrogenic potential, in accord with the origin of Rydberg states that we discussed in the introduction.

3.3. Analytical analysis of the molecular ion core model

Perturbation theory [40] has proven successful in calculating some properties of Rydberg molecules, such as spatial confinement effects [41]. Here, we apply first order time-independent perturbation theory to investigate the energy shift of the Rydberg electron bound to the spherical potential relative to that of hydrogen [35].

$$E_{nl} = E_n^0 + \langle \psi_{nl} \mid \Delta V \mid \psi_{nl} \rangle_{r \leqslant \sigma} \tag{3.6}$$

Here

- E_n^0 is the energy of the electron in a hydrogen atom with principal quantum number n;
- ψ_{nl} is the radial wavefunction of a hydrogen atom with principal quantum number n and orbital quantum number l;
- ΔV is the difference between the model potential from Eq. (3.5) and the hydrogenic potential (Coulomb potential). For $r > \sigma$, this potential difference is zero;

The integrals in Eq. (3.6) are evaluated analytically for 2s, 2p, 3s and 3p orbitals.

3.4. Numerical analysis

The Schrödinger equation for the molecular ion core model is solved using COMSOL Multiphysics Software® (Version 3.5). The Dirichlet boundary is a spherical box with a radius of $50a_0$ (a_0 = 1 Bohr Radius ≈ 52.9 pm). To achieve higher accuracy in a small spatial range around the origin, an algorithmically optimized rectangular block of $17a_0{*}17a_0{*}13a_0$ is also pre-set symmetrically about the origin. The electric potential resolution is 6 points per Bohr Radius in each Cartesian direction inside the block and 1 point per Bohr Radius for the other parts in the sphere. The whole system

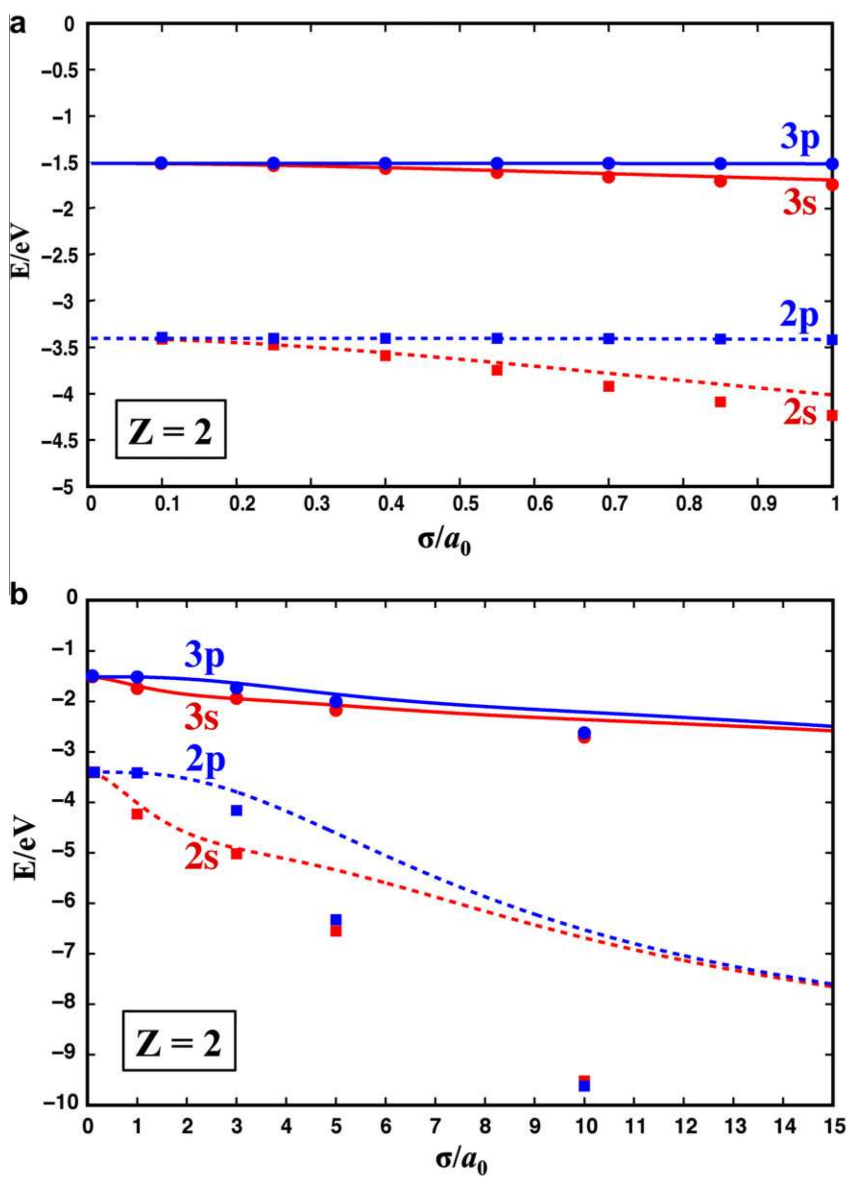

Fig. 4. Rydberg electron energies of the molecular ion core model in which $Z = 2$. (a) Sphere radius σ ranges from 0 to $1a_0$. (b) Sphere radius σ ranges from 0 to $15a_0$. The curves are the results obtained from perturbation theory and the scatter points are obtained from the numerical-grid method.

is then meshed with "finer" settings in the software package. The geometry and mesh grids are shown in Fig. 3.

With the system mesh grids and potentials defined, we utilize the PARDISO linear solver [42] from the software package to solve for the eigenvalues of the Schrödinger equation. Parallel calculations are accomplished using an 8-core CPU and 4 GB RAM on an SGI Altrix XE 1300 cluster. Electron energy level results are shown in an eigenvalue list and the orbitals can be plotted directly in the software GUI.

4. Results and discussion

4.1. The spherical ion model: analytical and numerical results

To analyze the Rydberg electron energy levels from the spherical model potential we first consider a central charge of $Z = +2$. The perturbation-theory Rydberg electron energies in the 2s, 2p, 3s and 3p states are plotted as a function of the molecular core radius σ, together with calculation results from the numerical-grid method for certain values of σ. As is indicated in Fig. 4, the electron energy level spectrum shows an order of $E_{3p} > E_{3s} > E_{2p} > E_{2s}$, which is in accordance with what we expect from quantum defect behavior as implied by Eq. (1.1) as well as by experiment data [14].

It is worth noting that when σ goes to zero the energy spectrum converges to $E = -1.51$ eV $= (-1/9)R_h$ for 3s and 3p orbitals and to -3.40 eV $= (-1/4)R_h$ for 2s and 2p. Physically, $\sigma = 0$ means the molecular ion core "shrinks" to a point, so that it just becomes the hydrogen atom (with the corresponding limiting electron energies -3.40 eV for $n = 2$ and -1.51 eV for $n = 3$).

Comparison of the perturbation theory and the numerical-grid results in Fig. 4(a) shows that the data from two methods are in very good agreement for small values of the core size. This indicates that the accuracy of the numerical-grid calculation is quite satisfactory, proving the validity of the numerical-grid calculation of the Rydberg electron energies. However, when the value of σ becomes large, the perturbation theory curves deviate from the numerical-grid results (Fig. 4(b)). The deviation at large sphere radii is understandable given the fact that the wavefunctions in the core begin to differ noticeably from that of a hydrogen atom. When that happens, first order perturbation theory might no longer provide a good approximation to the energy levels.

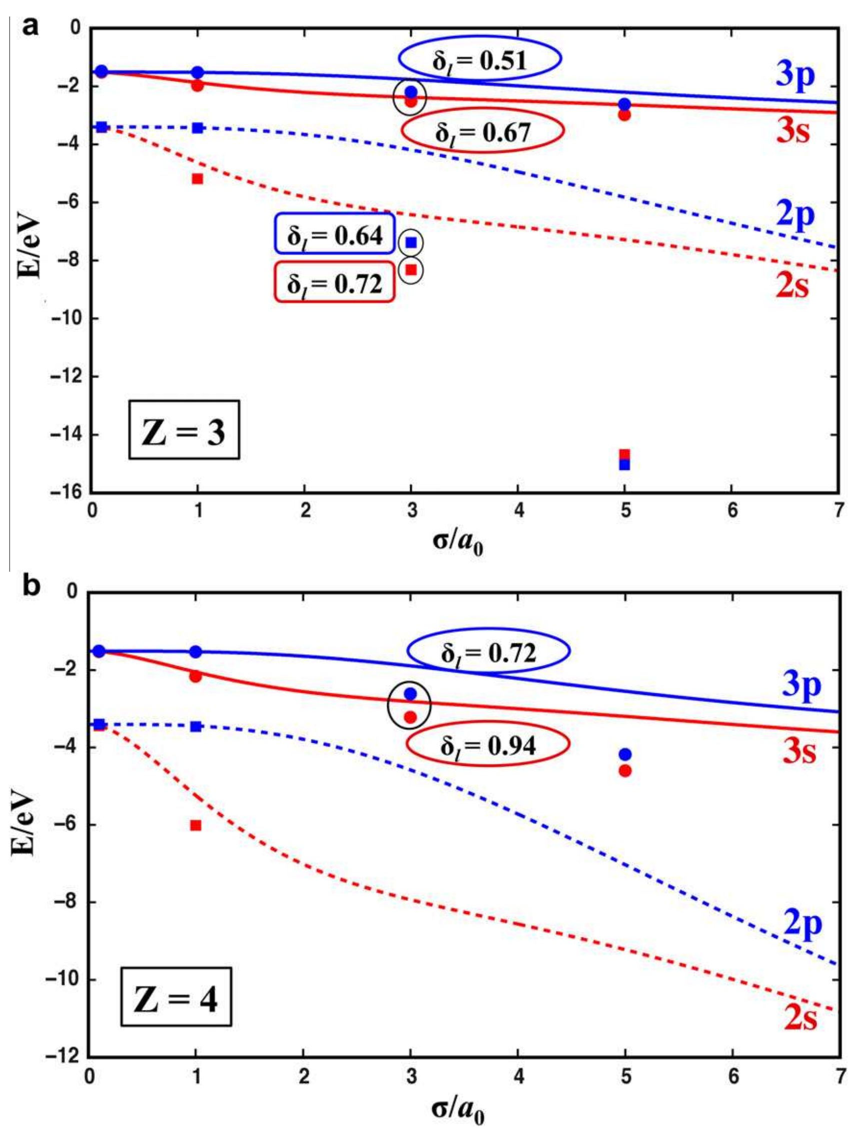

Fig. 5. Rydberg electron energies of the ion core model. Some quantum defect δ_l values calculated from the numerical-grid results are also shown. (a) $Z = 3$. (b) $Z = 4$. The curves are the results obtained from perturbation theory and the scatter points are obtained from the numerical-grid method.

As the central charge Z of the model potential increases, say to $Z = 3$ or 4, we observe a larger gap between perturbation and numerical-grid results, Fig. 5. It is reasonable to suggest that perturbation theory becomes worse in these cases since larger Z will naturally induce a larger difference between the Rydberg electron wavefunction and the hydrogenic wavefunction.

The main point, however, is that the results of Figs. 4 and 5 bear out the structure sensitivity of the Rydberg electron energy: the Rydberg electron energy depends strongly on the single structural parameter of the model, namely the σ.

It is instructive to calculate the quantum defect δ_l with the Rydberg electron energies calculated from numerical-grid method. From Figs. 4 and 5, it is clearly seen that for all the orbitals, increasing either σ or Z value in the model potential leads to more negative Rydberg electron eigenvalues, hence larger δ_l values. Potentials with $Z = 3$ and $\sigma = 3$ (Fig. 5(a)) as well as $Z = 4$ and $\sigma = 3$ (Fig. 5(b)) show a δ_l of 0.7–0.9 for s orbitals and 0.5–0.7 for p orbitals. These values fall well into the range of the empirical quantum defect data.

4.2. Comparison to tertiary amines

We further investigate the δ_l values from some experiments, and compare with our results from numerical-grid method. The Rydberg Fingerprint spectra of trimethylamine (TMA) and tripropylamine (TPA) are shown in Fig. 6. The Rydberg electron energies of the 3s and 3p orbitals are obtained from Fig. 6 and the corresponding δ_l values are calculated by Eq. (1.1), as summarized in Table 1. The numerical-grid calculated δ_l values shown in Fig. 5 are in reasonable accordance with the experimental values.

With its larger aliphatic hydrocarbon chains, TPA has a larger molecular radius than TMA. Yet the δ_l values of the same orbitals in TPA are smaller than those in TMA, apparently contradicting what we observed in our molecular ion core model when we increase the core radius σ. This is probably because although TMA and TPA molecules have similar molecular geometries, the contrast in the molecular size is not the only difference; other molecular properties, such as molecular potential (our Z parameter), could also be quite different. However, when we look at the change of the δ_l value from TMA to TPA for 3s and 3p orbitals respectively, we note that the decrease of the quantum defect of the 3s orbital is larger than that of the 3p orbital. The physical reason for this trend – which the model correctly captures – is that a Rydberg electron in a 3p state has a much smaller probability to appear within the molecular ion core than it does in a 3s state. Hence the 3p Rydberg electron is less perturbed by (and scattered from) the molecular ion core, resulting in the noted relative size dependence of its quantum defect values.

While some of our numerical-calculated δ_l values are in accordance with the experimental data, the inconsistency in the δ_l value variation trend we discussed above encourages us to further investigate geometric models, to better understand the model parameters, and in particular to develop an independent atom model for Rydberg states.

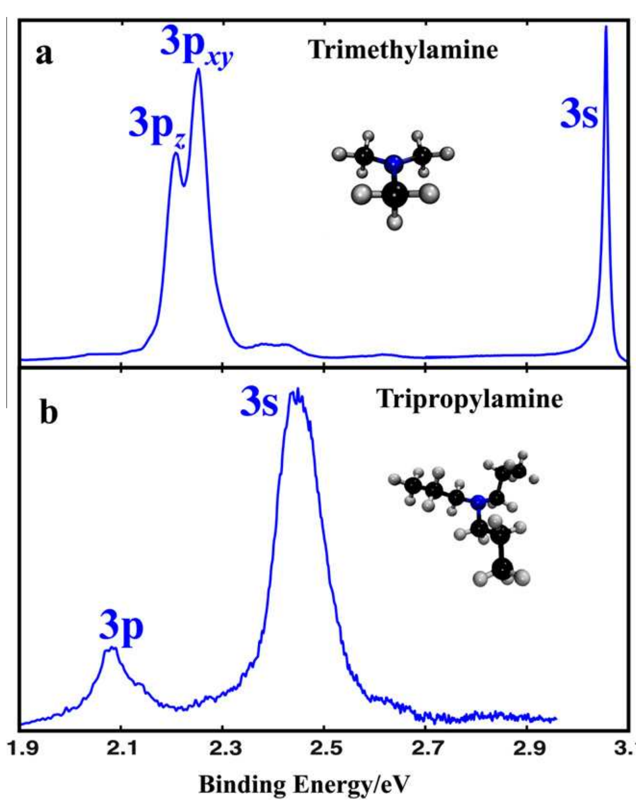

Fig. 6. Rydberg Fingerprint spectra of some tertiary amines: (a) trimethylamine (TMA); (b) tripropylamine (TPA). Both spectra are taken by exciting the molecules, seeded in a molecular beam, to the Rydberg states using laser pulses near 208 nm, and ionizing with pulses at 416 nm.

Table 1
Summary of the Rydberg electron energies and the corresponding quantum defect values of TMA [43] and TPA (this work).

	3s		$3p_{x,\,y}$		$3p_z$	
	E (eV)	δ_l	E (eV)	δ_l	E (eV)	δ_l
TMA	−3.087	0.90	−2.251	0.54	−2.204	0.52
TPA	−2.45	0.64	−2.08	0.44	−2.08	0.44

5. Closing remarks

While Rydberg Fingerprint Spectroscopy has been utilized extensively as an effective structural spectroscopy method with ultrafast time resolution, a detailed understanding of the structure sensitivity as well as the dependence of the energy on molecular parameters remains to be achieved. Our presented work uses a quantum mechanical analysis as well as numerical computations to explore a model potential that approximates molecular systems and that provides insights into the structure dependence of RFS.

Similar to electron diffraction, the structure sensitivity of RFS originates from the phase shift of the electron wavefunction during its scattering from the positively charged atomic nuclei and the surrounding electrons. The energy of a Rydberg electron in RFS and the scattering amplitude in electron diffraction share the same mathematical form; the representation of the potential operator $\hat{V}$ and the scattering operator $\hat{L}$ both reflect the interaction of the probing electron with the charge distribution of the molecule.

The application of perturbation theory and the numerical-grid method to the molecular ion core model with a potential described in Eq. (3.5) reveals the dependence of Rydberg electron binding energies on molecular radius σ. While the physical meaning of the parameter Z in this model remains unclear, we do obtain quantum defect values that fall well into the range of the empirical and experimental data for certain (Z, σ) combinations, although the δ_l value variation trends at constant Z are inconsistent with the experimental data. Encouraged by these results, we will continue to optimize the molecular ion core model parameters in order to reach a better understanding of the meaning of Z. We note as well, that the agreement between the analytical and numerical calculations with the model potential proves the utility of the numerical-grid method in solving for molecular Rydberg electron energies. Nonetheless, it will clearly be important to introduce more realistic molecular features into the model in order to begin to quantitatively reproduce the molecular spectra.

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